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Machine Learning–Based Prediction of Post-Harvest Loss Risk for Smallholder Agribusinesses in Developing Countries

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Abstract

Losses after harvest have always been a significant issue in developing states that lowers the income of farmers and a restriction of food availability. This is more so in smallholder agribusinesses as a result of poor storage conditions, low practices in handling, and the lack of access to affordable technologies. The goal of this paper is to come up with a machine learning-based model of predicting the risk of post-harvest loss based on low-cost and the availability of data.

The research design was the quantitative and experimental study based on the structured information used in the Nigeria General Household Survey [World Bank, 2024]. The use of important variables in it were the type of crop, manner of storage, length of storage, geographical area and the alleged causes of loss. Three machine learning models which include a Logistic Regression model, a Decision Tree, and a Random Forest were implemented and tested on the basis of accuracy, precision, recall, F1-score, and AUC-ROC values.

The results show that the most important predictors of post-harvest losses are storage method and storage duration, with the type of crop and location coming next. Random Forest was the best predictive model and showed good capability to fit intricate data associations. Decision tree models were also feasible as well as interpretable, while Logistic Regression served as a strong baseline. As was shown in this paper, even the simplest models of machine learning can be applied to make good predictions on whether there would be losses after harvest and issue correct tips to smallholder farmers. The projections will be meant to provide an excellent guide toward improving decision making in the storage, handling and marketing practices. The results of the study demonstrate the feasibility of low-cost decision support systems and the possibility of creating data-driven solutions to improve productivity, minimise losses, and promote food security in developing countries.

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Chapter 1

Introduction

1.1 Background and Context

Losses during harvest are still one of the severe issues in the international food systems in developing nations where the harvests efficiency has been significantly limited by the defects of the supplying channel that have reportedly resulted in the shortage of food stocks and crop harvests. With every harvest, a high percentage of crop products is not utilized; the level of wastage is higher in places like Sub-Saharan Africa and South Asia due to lack of infrastructure and lack of technology. These losses also weaken the economic sustenance of smallholder farm structures besides depleting food security.

Farmer income is especially affected by the effects of post-harvest losses. Smallholder farmers, who form a significant portion of agricultural producers in developing economies, tend to be financially vulnerable. Losses realized after harvest directly reduce marketable output and decrease profits. Research has indicated that handicapping in after-harvest processing and storage considerably reduces economic returns, contributing to poverty cycles amongst agricultural populations [Opara et al., 2025]. Uncertain losses also cause a rise in uncertainty, making it difficult for farmers to make sound production and investment decisions.

Another big issue associated with post-harvest losses is food security. With world hunger on an increasing curve, curtailing loss is an option to enhance food supply without necessarily increasing food production. [Fadiji et al., 2023] note that sustainable realization of food systems requires the reduction of post-harvest losses in areas with already existing food shortages. It is particularly troubling that resources spent on perishable goods that are not consumed result in a loss of access to nutritious food as well as dietary deficits.

In many developing countries, supply chains tend to be inefficient and often fragmented, which increases post-harvest losses. Inadequate transport systems, inefficient storage, and lack of coordination between parties add to delays and rotting of products. In wholesale markets, fresh produce is wasted in high quantities as a result of poor demand prediction and management methods [Opara et al., 2025]. This underlines the necessity for better coordination and the use of data-driven decision making across the supply chain.

The latest developments in artificial intelligence and machine learning present an opportunity to tackle such issues. Predictive models have the ability to identify trends in environmental and operating data that can predict any losses and help in making proactive decisions. The implementation of neural network-based strategies to forecast post-harvest losses at production and market scales has shown the potential of data-driven tools in enhancing efficiency [Guo et al., 2025].

Likewise, predictive modeling in the garri production demonstrated that machine learning can detect the main sources of losses and assist in targeted changes [Adewumi et al., 2025].

Although these advances have been made, the uptake of these technologies is still low in smallholder situations because of financial, technical, and data availability constraints. This highlights why basic, easy-to-reach machine learning solutions suited to the realities of the developing world are necessary, which underpins the current research.

1.2 Problem Statement

In developing nations, post-harvest losses are still a major problem, with very high percentages of agricultural products being wasted during storage, handling, and distribution. Research shows that losses of up to 30 per cent are possible, especially in low-resource environments where management practices and infrastructure are subpar [Affognon et al., 2015]. The net effect of these continuous losses is a weakening of food security, decreased incomes of farmers, and a curtailment of agricultural systems on the whole.

Despite the potential of technological changes through artificial intelligence and machine learning to overcome post-harvest inefficiencies, current solutions are not accessible to smallholder farmers. Several of these systems demand sophisticated digital infrastructure, high-quality data, and technical experience that are not suitable in resource-constrained settings. Moreover, such a transformation in rural agricultural settings is not supported financially and is hampered by a lack of digital literacy [Wolfert et al., 2017].

Another equally important constraint is that the majority of currently available tools are geared towards large-scale commercial farming with little regard given to the realities of running a smallholder agribusiness. Consequently, this leads to an evident gap between technological solutions and end-user needs. Also, existing methods tend to be either monitoring or control-oriented, which restricts their ability to be proactive in preventing loss.

Therefore, there is an urgent need to develop simple, cheap, and interpretable predictive instruments based on easily accessible data to forecast the risk of post-harvest losses. It is important to bridge this gap and ensure that smallholder agricultural systems are resilient and profitable through data-driven decision-making.

1.3 Research Aim, Objectives and Questions

1.3.1 Aim

The aim of this study is to identify and validate machine learning models capable of predicting post-harvest loss risk using low-cost data.

1.3.2 Research Objectives

- To determine and assess the storage, transportation, handling, and environmental conditions that contribute most to post-harvest losses.
- To assess the predictive accuracy of simple machine learning models by using sparse data to predict the post-harvest loss risk for smallholder crops.
- To evaluate the ways machine learning forecasts can be used to aid realistic agribusiness management choices.

1.3.3 Research Questions

1. What storage, transportation, handling, and environmental factors have the greatest effect on post-harvest losses?
2. To what extent can simple machine learning models correctly indicate whether a post-harvest loss is going to occur based on only limited data?
3. What can these projections do to facilitate business management of agriculture?

1.4 Significance of the Study

The research is significant at the academic level, the practice level, and at the policy level. Academically, it promotes the use of machine learning in agricultural decision support systems, especially in the under-researched field of post-harvest loss prediction. The study addresses the gap between theoretical machine learning methods and practical agricultural problems by emphasizing simple and understandable models.

In practice, the study provides a cheap and readily available solution to smallholder agribusinesses. It offers a tool to help farmers predict the presence of post-harvest risks and make informed decisions about storage, transportation, and handling by using readily available data and computationally efficient models. This assists in minimizing losses and maximizing profitability.

In policymaking, the findings promote long-term initiatives regarding improvement in food security in developing nations. The study is strategic to long-term agricultural development and the efficient use of resources, since it offers the capacity to employ data to conduct interventions and enhance efficiency across the supply chain.

1.5 Scope and Limitations

The paper focuses on the development of a prototype machine learning model to forecast the hazard of post-harvest losses in a smallholder agribusiness in the developing world. It is not a complete research of all farm produce but is narrowed down to a small number of farm crops so that it can be manageable and allow for an in-depth study.

The research involves simple, structured tabular data and does not involve complicated methods like deep learning because such methods require large amounts of data and high computational power, which are not available in low-resource settings. As an alternative, the focus is directed toward interpretable and inexpensive models that can be implemented in practice.

There are a few limitations that are acknowledged. The data will be relatively small, potentially impacting the model's predictability and generalisability. Also, the research is limited to certain geographical and operational settings, which can restrict the extrapolation of results to other areas. Being a prototype, the system is not implemented in real-time situations.

1.6 Dissertation Structure

There are six major chapters in this dissertation. Chapter 1 presents the research topic and problem, and outlines the research objectives and questions. Chapter 2 provides a review of literature in the field of post-harvest losses and the application of machine learning in agriculture. Chapter 3 covers machine learning models for prediction, decision support systems, the research gap, and the conceptual framework. Chapter 4 describes the research methodology, including

data collection, model choice, and data evaluation procedures. Chapter 5 reports the results and discussion of the machine learning models. Lastly, Chapter 7 sums up the research, outlines the contributions, and offers future research proposals.

Chapter 2

Literature Review

2.1 Introduction

This chapter examines literature that is available about post-harvest losses and machine learning in agriculture. It looks at the major concepts, theories, and empirical evidence that inform the research, especially on aspects that affect post-harvest losses, predictive modeling methods, and decision support systems. The purpose of the review is to formulate a clear theoretical background through the synthesis of existing knowledge and to fill gaps that exist in the literature. With the critical review of past literature, this chapter underscores that there is a need for straightforward, data-based solutions that suit the agribusinesses of smallholders in developing nations, and that is why the current research has a factor to play in the field.

2.2 Post-Harvest Losses in Developing Countries

Post-harvest losses may be defined as the increase or decrease in an agricultural product in terms of amount and quality between the time of harvest and the time of consumption. These losses include physical loss, nutritional loss, and loss in market value, especially for perishable commodities like fruits and vegetables. In the developing world, infrastructural, technological, and managerial deficiencies are particularly acute, and the extent to which the value chain is prone to post-harvest losses may range from 30–40 per cent of total output [Rajapaksha et al., 2021].

Post-harvest losses may be divided into two subcategories: quantitative losses, which include weight or volume loss, and qualitative losses, which include loss in texture, taste, nutritional value, and appearance. Although quantitative losses cause a direct decrease in supply, qualitative losses typically affect market price or result in rejection, thus exerting a multiplied economic effect on smallholder farmers [Wakene and Sharew, 2024].

There are a number of associated factors that cause post-harvest losses in developing countries, and one of the most important is storage conditions. Poor storage facilities such as refrigeration problems or poor ventilation hasten spoilage and the growth of microbes. For example, insufficient storage of fruits and vegetables in tropical conditions contributes to the rapid physiological degradation of the product due to high respiration rates and moisture loss [Rajapaksha et al., 2021]. On the same note, most smallholder farmers usually use ancient forms of storage which cannot sustain the quality of products over time.

Post-harvest turnover is also increased by transportation and logistical inefficiencies. Mechanical damage and spoilage are caused by poor road systems, extensive transportation times, and the absence of temperature-regulated transport systems. Rice producers in Liberia

face lower profitability partly because they lose profits in transportation and after harvesting [Pluato et al., 2024]. Loading, unloading, and transit operations may cause physical damage that decreases the shelf life and marketing ability of produce considerably.

Handling practices are also very significant in defining outcomes after harvest. Poor agricultural practices such as insufficient knowledge of the right harvesting methods, inappropriate packaging, and rough handling exert more vulnerability to bruising and contamination. Research conducted on cucumber crop production has indicated that inappropriate handling practices and minimal access to training by small-scale farmers are primary contributors to post-harvest losses [Ukwuaba et al., 2024]. The absence of awareness on best practices for post-harvest management usually adds to these concerns.

The issue is also exacerbated by environmental and climatic conditions. Microbial growth and enzyme activity are favored by the presence of high temperatures and humidity levels, as is the case in most developing areas, thereby promoting faster spoilage. Climatic factors have a significant effect on tomato losses in Ethiopia, especially where storage and cooling facilities are unavailable during peak harvesting seasons [Wakene and Sharew, 2024]. Climate variability also interferes with supply chains, making it difficult to maintain consistent quality after harvest.

The post-harvest losses have significant economic and social impacts. The economic implications are that losses cause farmers to have less income and profitability, resulting in their inability to invest in better technologies or practices. In Liberia, smallholder farmers are an example of how a major source of lost livelihoods is the inefficiencies that occur during the post-harvest phase [Pluato et al., 2024]. Socially, post-harvest losses are known to contribute to food insecurity due to the lack of healthy food in local markets. Furthermore, when there are inefficiencies in supply chains, food prices rise, burdening the poor population disproportionately.

Moreover, post-harvest losses compromise overall agricultural development objectives by limiting food system efficiency. The losses that happen at different stages reduce economic returns and food availability, as studies on vegetable supply chains such as amaranthus show [Oluoch et al., 2024]. It is thus important to address these issues not only to improve farmer livelihoods but also to increase food security and sustainability in developing countries.

2.3 Factors Influencing Post-Harvest Loss

The intricate interaction of storage, transportation, handling, and environmental factors is the cause of post-harvest losses in developing countries. These determinants must be understood in order to identify key predictors of loss risk, which directly answers Research Question 1 of this study.

2.3.1 Storage Conditions

Storage is very important in the maintenance of quality and shelf life of agricultural produce. Smallholder farmers in most developing nations do not have access to proper materials like cold rooms or controlled atmosphere facilities to store their goods. This causes fruits and vegetables to be kept in ambient temperatures, thereby increasing spoilage. According to [Rajapaksha et al., 2021], insufficient storage facilities such as improper ventilation and temperature control are an important cause of fruit and vegetable spoilage. In the same manner, [Wakene and Sharew, 2024] mention that tomatoes stored in poor conditions are actively degraded physiologically, causing quantitative and qualitative losses. Lack of affordable storage technologies thus continues to be a significant limitation towards minimizing post-harvest losses.

2.3.2 Transportation Logistics

Another significant cause of post-harvest losses is transportation inefficiency. Lack of appropriate road networks, long distances to travel, and the absence of refrigerated transportation systems subject produce to mechanical damage and unfavorable environmental conditions. [Pluato et al., 2024] show that in Liberia, high losses in transportation significantly decrease the profitability of rice production among smallholder farmers. Any failure to transport produce to markets promptly raises the chances of spoilage, mainly for perishable goods. Also, poor packaging during delivery will usually lead to compression and bruising, consequently lowering product quality and market value.

2.3.3 Handling Practices

The management of practices at different parts of the supply chain, such as harvesting, sorting, and packaging, has a significant impact on post-harvest results. Misuse of handling techniques may result in physical injuries to produce, making it more vulnerable to microbial infection and rapid breakdown. [Ukwuaba et al., 2024] identify smallholder cucumber producers, whose major problems are improper harvesting practises, rough handling, and non-standardised packaging, as primary contributors to post-harvest losses. Such obstacles are usually associated with inadequate training of farmers and ignorance of optimal farming practices.

2.3.4 Environmental Variables

Conditions in the environment, especially temperature and humidity, are decisive factors of post-harvest loss. Hot weather increases the rate of respiration in produce, resulting in quicker ripening and ultimate spoilage, while elevated moisture content encourages the development of microorganisms. [Wakene and Sharew, 2024] note that losses of tomatoes in Ethiopia are greatly caused by fluctuations in temperature and humidity. Similarly, [Rajapaksha et al., 2021] observe that climate conditions in tropical regions contribute to increasing post-harvest degradation because there are no mechanisms of temperature control. Seasonal differences also affect the pattern of losses, where peak harvest seasons often coincide with weather conditions that favor spoilage.

2.3.5 Cohesive Effect of the Supply Chain

These factors rarely work in isolation; they interact at various stages throughout the supply chain, increasing aggregate losses. [Oluoch et al., 2024] explain that in the amaranthus supply chain, losses are experienced during harvesting and distribution to the market due to the triple interaction of poor storage, unproductive transport, and inefficient handling. This demonstrates the importance of holistic approaches to post-harvest management.

Overall, post-harvest losses in developing countries are mostly affected by storage conditions, logistics involved in transportation, post-harvest handling behaviors, and environmental factors. The identification and quantification of these variables are essential to establishing predictive models that benefit smallholder agribusinesses through data-driven decisions.

2.4 Machine Learning in Agriculture

Machine learning (ML) as an agricultural application has been on the rise in the past few years due to the necessity of obtaining better productivity, efficiency, and sustainability in food systems. ML methods make it possible to analyze large and intricate datasets, getting patterns and

making predictions to help guide decisions. In underdeveloped economies, where the availability of state-of-the-art technologies is curtailed by resource requirements, easy-to-understand and interpretable ML models offer a chance at improving agricultural activity, especially in yield forecasting and crop management.

Among the most outstanding uses of ML in farming, crop yield forecasting can be mentioned. Yield can be forecasted with sufficient accuracy to assist planning, manage the supply chain, and secure food. Conventional ways of estimating yields usually take into consideration historic averages or manual estimations that fail to account for dynamic environmental and operating variables. ML models on the other hand can combine various variables like weather, soil characteristics, and agricultural methods to give better predictions. As pointed out by [Jabed and Murad, 2024], machine learning methods, such as regression models and ensemble methods, have been shown to perform highly when predicting crop yields in various agricultural environments.

Along with predictive yielding, disease detection and crop health monitoring are other crucial areas of application of ML. The early detection of plant diseases is critical to minimizing losses and improving crops. The output of machine learning algorithms, especially those that work with image data, is highly accurate as the algorithms attempt to find patterns in leaf color, texture, and shape to detect disease symptoms. [Meghraoui et al., 2024] suggest that ML-based disease detection systems can offer timely responses, eliminating the necessity to conduct manual inspections or use expensive specialist expertise.

In addition to particular applications, the greater value of ML in agriculture lies in the fact that it can facilitate processes of data-driven decision-making. Pattern recognition is one of the major strengths of ML, in which an algorithm can establish trends and relationships among very large and complex datasets. This feature is particularly applicable to the agricultural setting, where numerous interacting characteristics can affect outcomes such as yield and post-harvest quality. According to [Jabed and Murad, 2024], this analytical capability has helped increase the accuracy and trustworthiness of agricultural forecasts.

Another relevant advantage of ML is its use in decision support systems. ML models have the ability to convert information into practical suggestions to allow farmers and agribusinesses to make good decisions. Predictive models can, for example, inform decisions regarding harvest timing, storage approach, and transportation logistics. [Meghraoui et al., 2024] suggest that the application of ML to the decision-making process in the agricultural domain will make the process more efficient and less prone to uncertainty, especially in resource-limited contexts. Notably, simple models such as logistic regression and decision trees can be used to make sure such systems remain interpretable and accessible to final users.

Although these are the benefits, there are still difficulties in the implementation of ML in the agricultural sector. Availability and quality of data are usually poor, especially in developing countries. Moreover, the complexity of models and their usability must be balanced, as very complex models might not be useful to smallholder farmers. However, the accumulating volume of literature shows that ML can achieve great changes in the field of agriculture by providing accurate predictions and enabling informed decision-making processes. In this research, these insights are applied to forecast post-harvest loss risk using machine learning methods.

Chapter 3

Related Work

3.1 Machine Learning Models for Prediction

The role of machine learning models in predictive analytics in the agricultural sphere is central when working with structured data under limited computing capabilities. In this paper, three popular and interpretable models are examined: Logistic Regression, Decision Trees, and Random Forests. They are suitable for smallholder situations and can be used with tabular farm statistics. These models are considered in the context of Research Question 2, which investigates how well simple machine learning methods can forecast post-harvest loss risk with limited data.

3.1.1 Logistic Regression

Logistic Regression is among the most basic classification algorithms applicable in machine learning. It is necessarily applicable to binary classification problems, such as identifying when the risk of post-harvest losses is high or low. The model uses input variables to estimate the probability of a specific outcome, mapping predictions in the range 0–1 by use of a logistic function. The simplicity and the ability to interpret results are among the greatest benefits of Logistic Regression. The features have a linear contribution to the prediction, which permits a clear understanding of how separate factors affect the result.

Logistic Regression has been applied widely in agriculture and is efficient with fewer computational requirements. As [Jabed and Murad, 2024] point out, less complex models like Logistic Regression are still very applicable in situations with small datasets where interpretability is paramount. This is more significant among the smallholders who need understandable and distinguishable decision support systems. However, an inconvenience of Logistic Regression is that it does not capture non-linear relationships of variables, and hence cannot describe the complexity of more intricate agroscenarios.

3.1.2 Decision Trees

Decision Trees offer a classification and prediction method that is rule-based and hence very intuitive and easy to interpret. This model effectively subdivides the dataset according to feature values, creating a tree-like structure of decisions. The internal nodes signify decision rules and the leaf nodes signify the predicted outcomes. This framework enables users to follow the decision-making process step by step, which proves to be important especially in real-world agriculture.

The use of Decision Trees can be accomplished effectively to work with both numerical and categorical data, as it is flexible enough to represent the agricultural datasets which frequently have different types of features. According to [Meghraoui et al., 2024], tree-based models have been extensively used in agricultural prediction activities because they are easy to interpret and can work with heterogeneous data. Moreover, Decision Trees are able to describe non-linear trends among variables and therefore have better prediction ability than linear models. They are however susceptible to overfitting, particularly in situations where the tree is overly complicated, which may lower generalisability to new data.

3.1.3 Random Forest

Random Forest is an ensemble learning technique which constructs several Decision Trees and aggregates their predictions to enhance overall performance. The aggregation of multiple tree outputs additionally reduces the risk of overfitting and enhances predictive performance. The trees in the forest are trained on random subsets of the data and features, which gives diversity in the models and makes them more robust.

Random Forest has consistently proven to outperform individual models in agricultural prediction work. According to [Jabed and Murad, 2024], ensemble methods, especially Random Forest, can be used to capture complicated interactions between variables while being relatively accurate. As [Meghraoui et al., 2024] further explain, these models can process a large feature set together with noisy data, both typical of an agricultural setting. Although, unlike single Decision Trees, a Random Forest cannot be readily interpreted, it has the ability to conduct feature importance analysis, which gives a picture of the most impactful variables in making predictions.

3.1.4 Relevance to the Study

These three models have been chosen based on trade-offs between the need to simplify, be interpretable, and be able to predict. Logistic Regression is transparent and easy to use, Decision Trees offer interpretable rule-based insights and higher accuracy, and Random Forest provides an enhanced ensemble learning prediction. Collectively, the models allow for a holistic analysis of post-harvest loss risk, even with a limited amount of data. Within the context of this comparative approach, Research Question 2 lies in evaluating the trade-offs between model complexity and predictive power in the smallholder agricultural environment.

3.2 Decision Support Systems in Agribusiness

Agribusiness decision support systems (DSS) refer to computer-based systems which help farmers and other agricultural stakeholders in coming up with sound decisions by analyzing data and providing actionable information. These systems are able to combine data of various sources, analyze through analytical models, and provide recommendations that aid in planning, resource allocation, and risk management. Regarding the case of smallholder agriculture, DSS can enhance productivity, alleviate uncertainty, and create sustainability through evidence-based decisions.

Having DSS in agriculture is essential because they can be used to analyze raw data and give meaningful information. The nature of agricultural decision-making is complicated because of the interplay of economic, operational, and environmental factors. This complexity can be reduced with the help of DSS that can detect patterns and predict outcomes and therefore facilitate timely and effective interventions. According to [Duan et al., 2021], DSS can help achieve sustainable agribusiness through enhancing efficiency, optimizing resource utilisation,

and long-term planning. Their multicriteria assessment shows how properly designed systems can help to improve both economic and environmental performance in farming environments.

In spite of these advantages, current DSS are usually limited, which restricts their use by smallholder farmers in developing countries. Many systems are technologically intricate and require good connectivity, high-quality data, and expert knowledge to operate. This poses a hindrance to the smallholder who might not have access to reliable internet, digital capabilities, or technical training. Moreover, some DSS target large commercial farming and are therefore less relevant to the conditions of small farming. [Duan et al., 2021] observe that technical performance is not the only factor to consider when evaluating the use of DSS: usability and accessibility are often disregarded in system design.

The other major weakness is the absence of contextual adaptation. Differences in agricultural conditions across regions are high, and DSS whose recommendations do not consider local practises, climate conditions, and the scarcity of resources can produce impractical and irrelevant recommendations. This enhances mistrust toward the system, granting it little usefulness to the end users.

These challenges mean that simple, low-cost, and understandable DSS that are smallholder-specific are urgently needed. These systems must be built on easily accessible data, employ clear-cut models, and contain clear and practical suggestions. This is the need that this study caters to, as it will create a machine learning-based predictive model which can assist in making decisions in the post-harvest process. Based on simplicity and usability, the proposed approach will fill the gap between sophisticated analytics and user intervention, which directly leads to Research Question 3.

3.3 Research Gap

Although machine learning has been extensively used in the agricultural sector, there are major gaps in its use for smallholder agribusinesses in developing nations. The existing literature has been largely concerned with high-accuracy types of models, including typically complicated algorithms and large-scale datasets, which may not be readily obtainable and applicable to smallholder settings [Jabed and Murad, 2024]. This creates a detachment between the use of technology and the reality of application.

A significant gap is the absence of cheap machine learning applications that are environmentally adapted to smallholder farmers. Most existing systems need powerful computing power, good internet connectivity, and specialist skills, which restrict their use in resource-limited environments. It is further noted that existing solutions are usually set to operate at large-scale commercial farming without much regard for the operational realities and constraints of smallholder agrobusinesses.

Another essential disparity is the underutilisation of basic, structured data. Although deep learning approaches have demonstrated good performance, they usually require large amounts of data, which is not frequently accessible in developing areas [Meghraoui et al., 2024]. Little literature has investigated how simpler machine learning models can successfully use small, tabular data to produce meaningful predictions.

In addition, the merging of predictive models and real actionable decisions is largely absent. Numerous reports are devoted to prediction accuracy without converting findings into practical suggestions for farmers. In this paper, these gaps are filled by identifying an interpretable, low-cost machine learning methodology on limited data to aid post-harvest decision making in smallholder settings.

3.4 Conceptual Framework

This paper follows a conceptual framework that relates input variables, machine learning processes, and decision-oriented outputs to respond to the risk of post-harvest losses. The framework is designed around three major components: inputs, processing, and outputs.

The **input stage** involves the environmental and operational factors that affect post-harvest results. These are variables in terms of temperature, humidity, storage method, storage duration, time of transportation, and handling measures. Such factors are generally known to be significant determinants of post-harvest losses in agricultural systems.

The **processing stage** includes the use of machine learning models, namely Decision Trees, Logistic Regression, and Random Forests. These models are used to predict relationships between input variables and historical data on losses, in order to determine trends and provide predictive information. They are chosen according to their capability to process structured information, their readability, and their low computational cost.

The **output stage** generates risk estimates and recommendations for action. These outputs aid decision making by allowing farmers to detect high-risk situations and implement proper post-harvest management strategies that mitigate losses and enhance profitability.

Chapter 4

Methodology

4.1 Research Design

The quantitative and experimental research design is followed in the present study because it will develop and test machine learning to forecast the post-harvest loss risk. A quantitative research method is selected because the given study aims at examining structured numerical data and determining relationships between the independent variables and the target outcome variable of the research. The experimental aspect involves the training and testing of various machine learning models to determine their predictive performance.

This design is appropriate because the research questions are objective in nature, measuring the relationship between factors that include storage conditions, transportation, and environmental variables and their effect on loss after harvest. The quantitative approach will make it possible to statistically assess results and compare individual models on objective grounds. Moreover, the experimental methodology facilitates reproducibility because models can be re-trained on exactly the same dataset and parameters. This design is hence suitable in the creation of a sound, data-driven decision support instrument for smallholder agribusinesses.

4.2 Research Onion Explanation

The research is grounded in the research onion model, which is conceptualized as a series of methodological choices layered from the outside in.

At the philosophical level, the research is based on **positivism**, as the researcher assumes that reality can be measured objectively using observable data. This is suitable because the variables included are measurable, such as temperature, conditions of storage, and time spent on transport.

The study uses a **deductive approach**, starting with the existing theory on post-harvest losses and machine learning, which is then tested using empirical data analysis. Hypotheses on the effect of certain factors and model performance are tested using statistical procedures.

The study design is **experimental** in the sense that the results of various machine learning algorithms are compared upon the same dataset to assess predictive accuracy.

The **time horizon** is cross-sectional, where data are analyzed at a particular point in time rather than over long periods. This is appropriate given the emphasis on forecasting loss risk depending on present conditions, as opposed to studying long-term trends.

4.3 Data Collection

This research uses secondary data as the main information source. Specifically, the data is sourced from the Nigeria General Household Survey (GHS-Panel Wave 5, 2023–2024) [World Bank, 2024], which is part of the World Bank's Living Standards Measurement Study – Integrated Surveys on Agriculture (LSMS-ISA). This survey provides organizational and country-based data concerning smallholder agriculture in Nigeria.

The data is organized and table-like, and hence machine learning applications can readily be applied to it. It includes both numerical and categorical variables involved in the post-harvest processes. The key features obtained include:

- Storage conditions (e.g. traditional versus improved storage methods)
- Time and distance of transportation
- Environmental factors (e.g. rainfall and temperature)
- Management of practices in harvest and post-harvest phases

The post-harvest loss risk target variable is calculated in accordance with available data based on the comparison of harvested and sold quantities, allowing each record to be classified as either high-loss or low-loss.

4.4 Data Preprocessing

Preprocessing of data is a very important measure to guarantee the size and quality of the data. The process begins with **data cleaning**, at which point inconsistencies, duplicates, and irrelevant variables are eliminated.

Missing values are handled using appropriate techniques such as imputation or elimination, based on the quantity and significance of the missing data. **Categorical variables** (e.g. type of storage and handling practises) are converted into numerical representation using encoding schemes like one-hot encoding or label encoding, to ensure compatibility with machine learning algorithms.

Numerical variables, wherever applicable, are standardized or normalized by adopting equal scaling of features. This is especially important for models that are sensitive to the magnitude of features, such as Logistic Regression. Preprocessing in general guarantees the appropriate organization, consistency, and aptness of the dataset for proper model training and evaluation.

4.5 Machine Learning Models

This study uses three machine learning models: Logistic Regression, Decision Trees, and Random Forests. Logistic Regression is an easy-to-understand model adopted as the base model. It works well when dealing with binary classification issues and gives explicit information on how the independent variables relate to the target outcome.

Decision Trees are included because of their rule-based structure, which makes the interpretation of decision paths easy. They are able to find non-linear correlations and can deal with both nominal and numerical data.

Random Forest is employed as an ensemble learning technique to enhance predictive performance. It minimizes overfitting by utilizing many decision trees and augments generalization. It is

applicable especially when identifying intricate interactions among variables.

These models have been selected because of their simplicity, interpretability, and ability to work with small to medium-sized datasets. They are less resource-intensive and easier to apply in a practical smallholder setting than deep learning models.

4.6 Model Evaluation Metrics

The models are tested based on four primary metrics: accuracy, precision, recall, and F1-score, supplemented by AUC-ROC.

Accuracy considers the total correctness of predictions. It can however be misleading when an imbalanced dataset is used. **Precision** measures the fraction of correctly predicted positive cases in reference to the total number of predicted positives. The importance of high precision lies in minimizing false alarms. **Recall** determines the model's capability to correctly identify actual positive cases. This is especially pertinent in this research as failure to identify high-risk situations of loss would result in poor decision-making. The **F1-score** balances precision and recall and is a good measure to use when data is imbalanced. The **AUC-ROC** (Area Under the Receiver Operating Characteristic Curve) measures the overall discriminative ability of the model across all classification thresholds. All of these metrics together give a complete analysis of model performance.

4.7 Feature Importance Analysis

An analysis of the influence of features on post-harvest loss risk is performed to identify the most influential factors. This is primarily accomplished through the Random Forest model, which provides a ranking of features according to their contribution to prediction accuracy. The analysis of feature importance allows the study to identify crucial drivers such as storage conditions, time of transportation, and environmental factors. This not only enhances the interpretability of the model but is also useful in making convenient decisions that show areas where interventions are likely to be most effective.

4.8 Code Repository

The complete source code, data preprocessing scripts, and model training pipelines developed for this project are publicly available in the following GitHub repository:

`https://github.com/achauque14/ML-Post-harvest-loss-risk-project`

The repository contains all Python scripts used for data cleaning, model training, evaluation, and visualization, along with instructions for reproducing the results presented in this dissertation.

4.9 Ethical Considerations

Ethical issues are put into consideration during the research process. The research involves the use of secondary data found in publicly accessible materials, and as a result, there is adherence to policies governing the use of such data. No personal identity information is released. Data privacy and confidentiality are ensured, and the analysis is conducted in a responsible manner in order to prevent improper use or misleading results. Also, the research

supports the responsible use of artificial intelligence, concentrating on the transparency and impartiality of model development.

4.10 Limitations of Methodology

This research is also methodologically limited in a number of aspects. Although the dataset used was adequate for the analysis, its size might limit the generalisability of findings. There is potential data bias (reporting error or regional bias) that may influence model performance. In addition, the use of cross-sectional data limits the temporal analysis of post-harvest losses. Lastly, the emphasis on simple models that are good in interpretability can lead to worse predictive power than more complex methods.

Chapter 5

Analysis and Results

5.1 Introduction

This chapter outlines the study analysis and results, which aim at identifying how machine learning models might be used to predict the post-harvest loss (PHL) risk of smallholder agribusinesses in Nigeria. Based on the Nigeria General Household Survey (GHS-Panel Wave 5, 2023–2024) [World Bank, 2024], the researchers estimate the behavior of post-harvest losses and determine the predictive power of the Logistic Regression, Decision Tree, and Random Forest models. The chapter covers exploratory data analysis, model evaluation, and important insights per the research questions.

5.2 Dataset Overview and Descriptive Analysis

This data consists of household-level agricultural data with variables such as the type of crop, storage methods, location, and the reported causes of post-harvest losses. After processing, the data was cleaned, missing values were imputed, and categorical values were encoded to facilitate analysis.

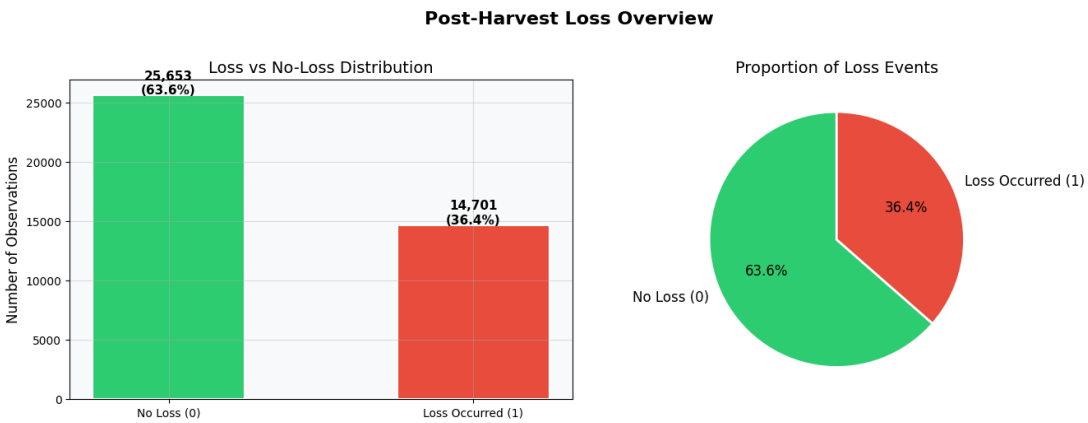

Figure 5.1: Post-Harvest Loss Overview

Figure 5.1 shows the total distribution of occurrences of post-harvest losses. The findings show that post-harvest losses were observed in some 36.4 per cent of observations. This indicates that despite the fact that some reporting areas experience more losses than others, a significant share of smallholder farmers is affected. The evident data imbalance further defends the

implementation of extra evaluation measures such as precision and recall, rather than relying on accuracy alone.

Figure 5.2 shows how post-harvest loss rates vary with crop type. There are crops which have very high rates of loss, like cabbage (more than 60 per cent), while other crops like orange and pawpaw show much lower loss rates. This difference is based on disparities in perishability, handling needs, and storage sensitivity. The results support the validity of post-harvest loss risk being sensitive to the type of crop, which directly answers Research Question 1.

The rates of losses in each geographic region and in rural and urban spheres are shown in Figure 5.3. The findings indicate a tendency for greater losses in the north than in the south. Furthermore, the rates of loss in rural areas are significantly higher (approximately 38 per cent) than in urban areas (approximately 26 per cent). This gives a hint that infrastructural constraints such as storage and mode of transportation have a great impact on the extent of post-harvest losses.

Reported causes of post-harvest losses are shown in Figure 5.4. The most common cause is pest and rodent infestation, followed by disease or mold and poor handling. Environmental conditions such as moisture and rainfall also play a role. These results point to the idea that biological and operational factors comprise the major causes of loss and therefore show the areas most susceptible to specific interventions.

The dependence between storage period and post-harvest loss is presented in Figure 5.5. Although the majority of storage times are of rather short character, the findings show that the longer the storage time, the higher the probability of loss. This implies that prolonged storage without proper preservation methods increases the chances of spoilage, underscoring the importance of proper storage management.

The correlation between the key variables is shown in Figure 5.6. Moderate positive relationships are observed between storage time, amount lost, and loss occurrence. Despite the weak correlation of most variables, this can be expected given the complicated nature of agricultural systems. The heatmap gives insight into variable relationships and aids in the selection of features for predictive modelling.

5.3 Model Performance Evaluation

Three machine learning models were assessed: Logistic Regression, Decision Tree, and Random Forest. They were divided into training and testing sets, and the results of model performance were evaluated in the form of accuracy, precision, recall, F1-score, and AUC-ROC.

Table 5.1: Model Performance Comparison

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC-ROC**
Logistic Regression	0.9742	1.0000	0.9293	0.9633	0.9715
Decision Tree	0.9905	1.0000	0.9738	0.9867	0.9920
Random Forest	0.9905	1.0000	0.9738	0.9867	0.9938

The findings in Table 5.1 show that both the Decision Tree and the Random Forest models reached a high level of performance, though the Random Forest achieved a higher AUC-ROC. Logistic Regression also showed good effectiveness but with lower recall, implying it was less sensitive to loss cases. These results prove that machine learning models may correctly predict the risk of post-harvest losses, responding to Research Question 2.

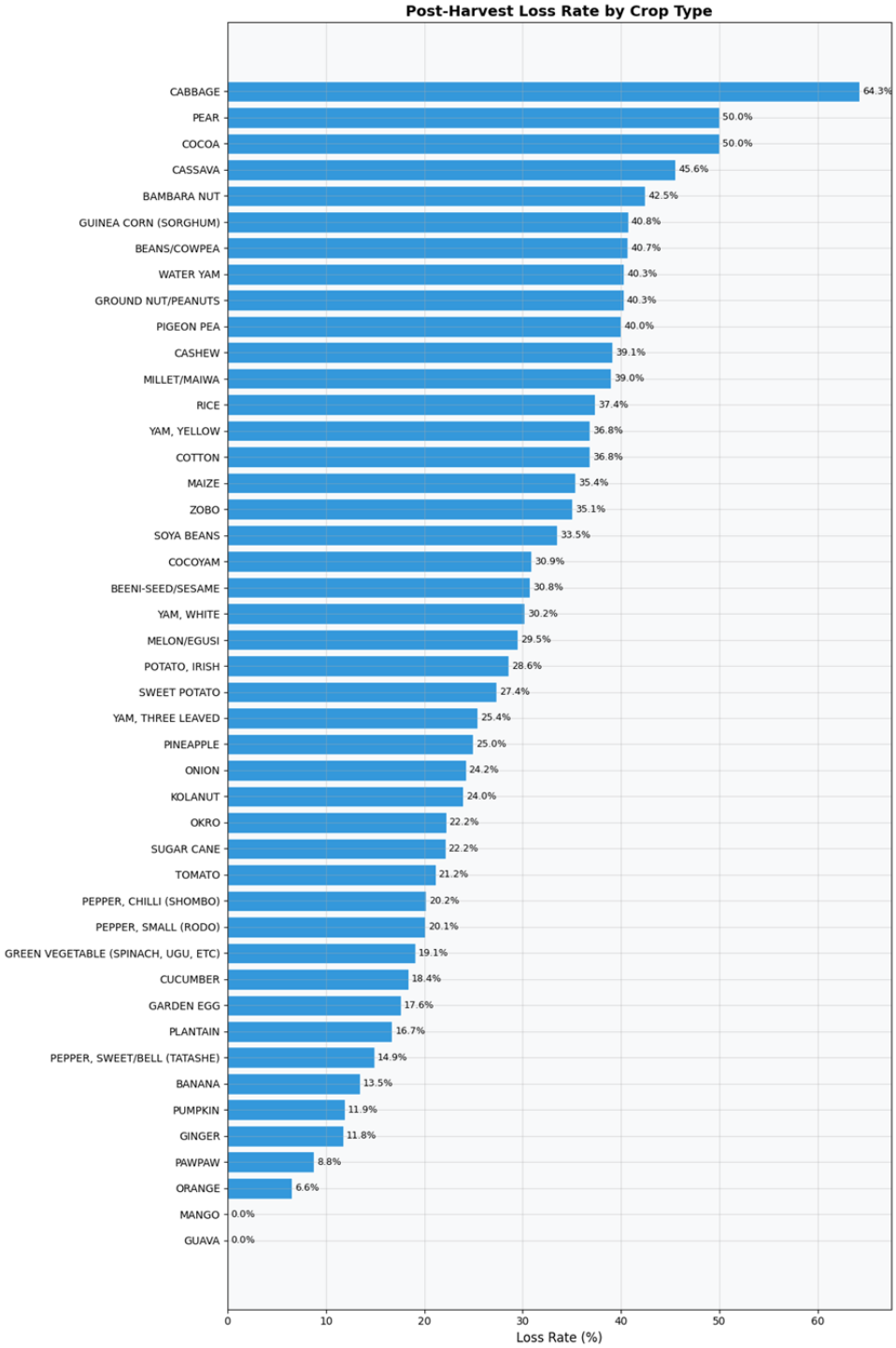

Figure 5.2: Post-Harvest Loss Rates by Crop Type

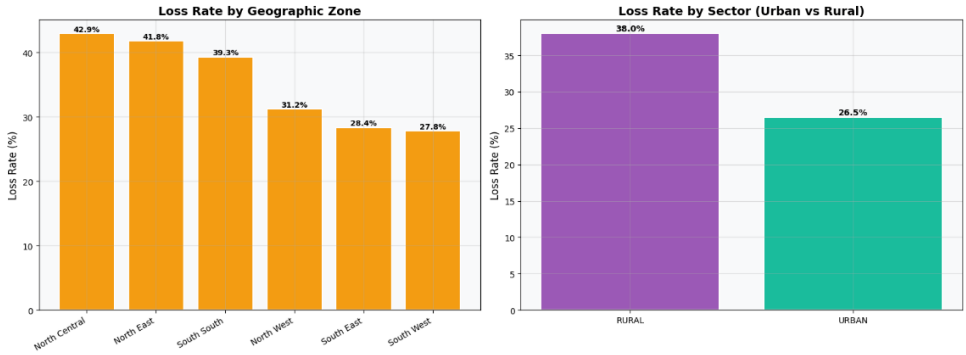

Figure 5.3: Post-Harvest Loss by Geographic Zone and Sector

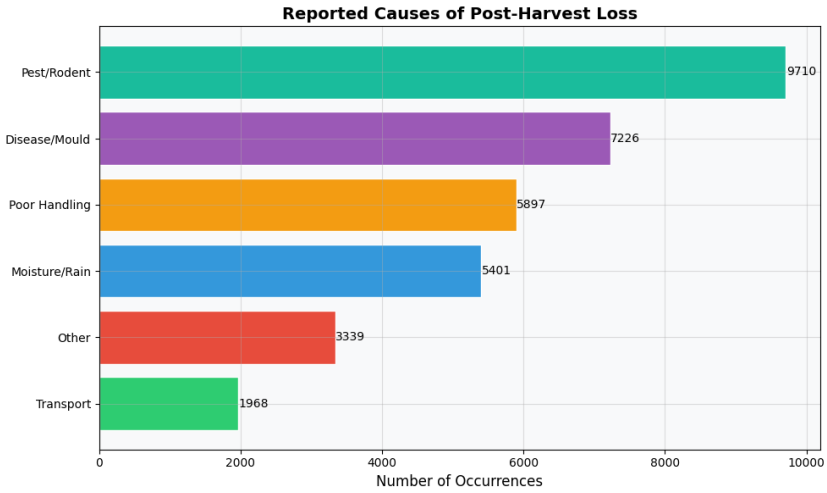

Figure 5.4: Reported Causes of Post-Harvest Loss

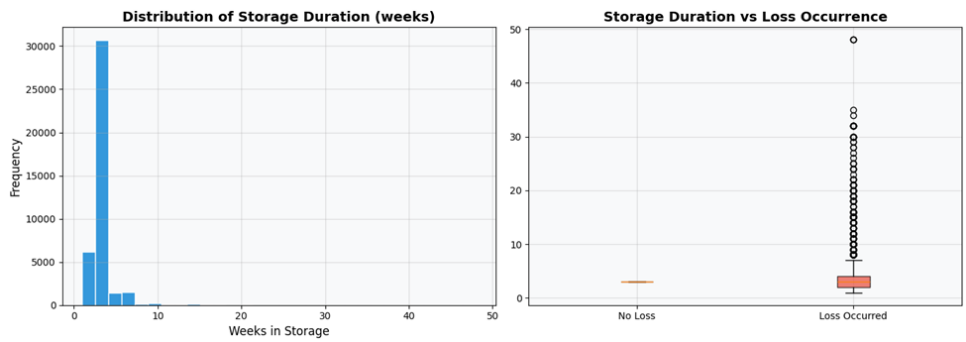

Figure 5.5: Storage Duration and Loss Occurrence

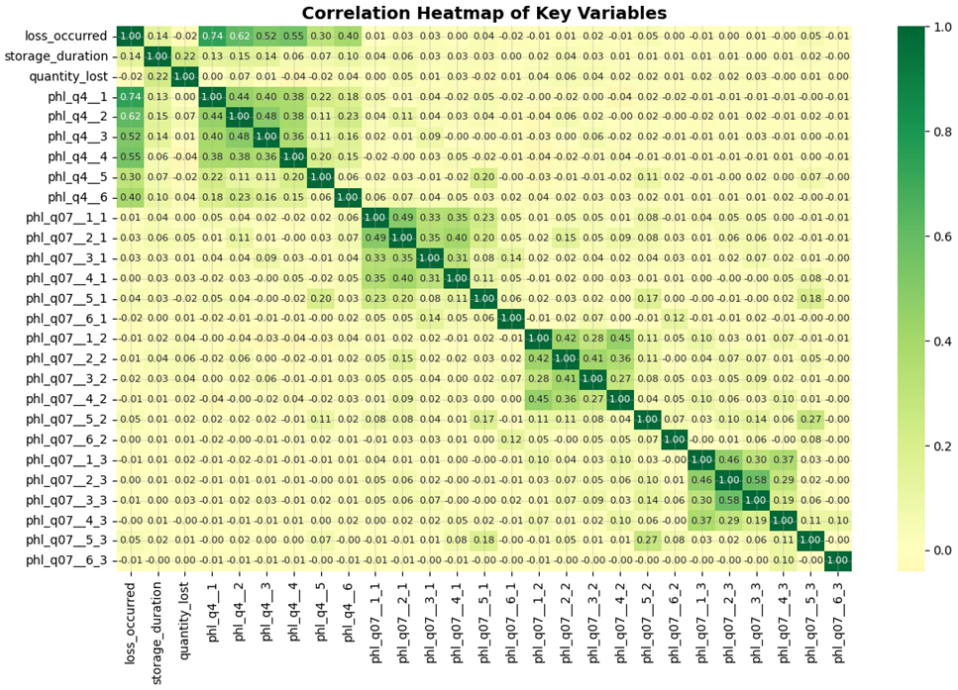

Figure 5.6: Correlation Heatmap of Key Variables

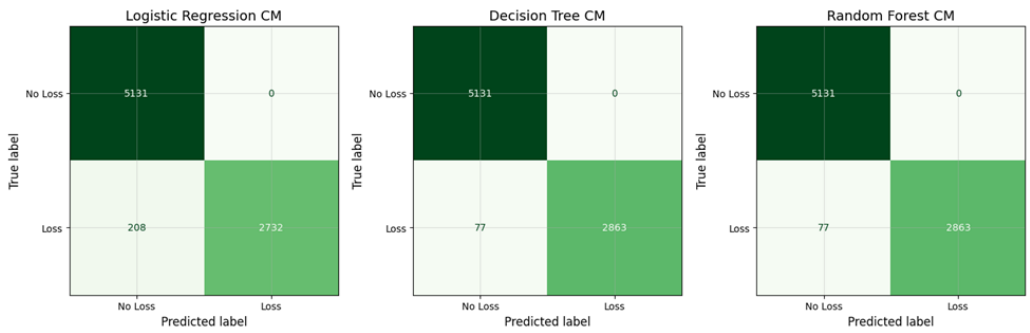

Figure 5.7: Confusion Matrices for Machine Learning Models

The confusion matrices are shown in Figure 5.7. Random Forest had the least number of misclassifications, confirming it as the strongest predictive model. The false negatives were greater in Logistic Regression, meaning it is strained when complex correlations are present. These findings also confirm the usefulness of ensemble methods.

5.4 Feature Importance Analysis

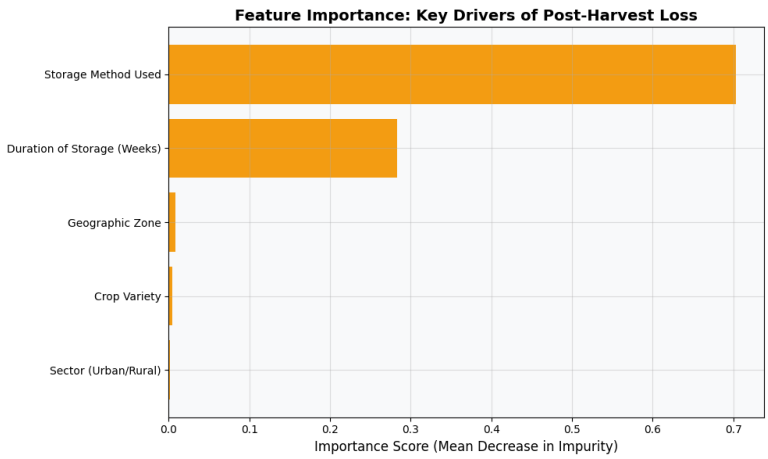

Figure 5.8: Feature Importance from Random Forest Model

Figure 5.8 demonstrates how features are important in predicting post-harvest loss. Storage method becomes the most determining variable, with storage duration coming next. Other variables, including geographic zone and crop type, are relatively less important. These results underline the significance of storage practice on loss risk and provide actionable guidance on how companies can minimise losses.

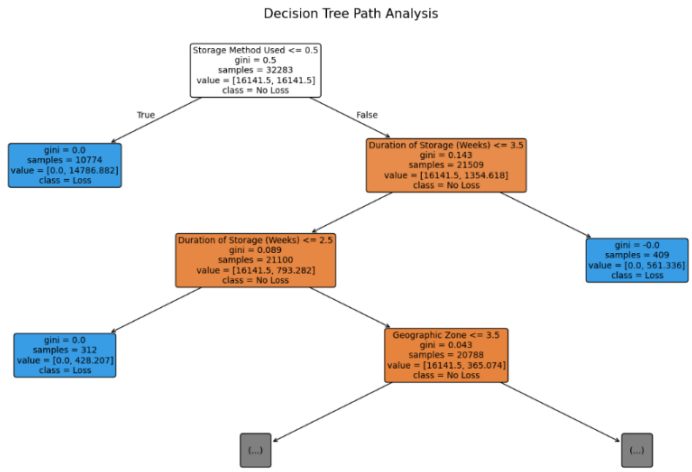

Figure 5.9: Decision Tree Path Analysis

Figure 5.9 shows the structure of the Decision Tree model. One of the key decision points identified in the model is storage method and storage duration, pointing to their significance in predicting outcomes. The uncovered tree may offer interpretable decision rules that can inform useful agrobusiness decisions, responding to Research Question 3.

5.5 Key Findings

The analysis yields a number of significant findings:

- A large number of smallholder farmers are impacted by post-harvest losses, with approximately 36.4 per cent of observations recording a loss event.
- The loss rates differ significantly depending on the type of crop, with perishable crops registering the highest loss rates.
- Distribution of losses depends on geographic and infrastructural location, with rural places being more affected than urban areas.
- The most significant predictors of post-harvest loss are storage method and storage duration.
- Among the models evaluated, Random Forest showed the best predictive performance as measured by AUC-ROC.

Chapter 6

Discussion

6.1 Introduction

This chapter critically reflects on the findings reported in Chapter 5 within the larger academic and practical framework of post-harvest loss management and machine learning in the agricultural industry. The research questions are used to organize the discussion by correlating empirical findings with the literature. It assesses the potential implications of the findings on theory and practice, the limitations of the study, and future research possibilities.

6.2 Discussion of Key Findings

6.2.1 Factors Influencing Post-Harvest Loss (RQ1)

This research has shown that the most influential factors determining post-harvest loss are the storage method, storage period, type of crop, and geographic location. Of these, storage-related variables proved to be most influential, as seen in the feature importance analysis and the decision tree structure.

This conclusion is very much in agreement with the literature. According to [Rajapaksha et al., 2021], poor storage systems contribute more significantly to post-harvest losses in developing nations, especially for perishable goods. In the same vein, [Wakene and Sharew, 2024] point out that inappropriate storage leads to faster spoilage, as is the case with most environmentally sensitive crops like tomatoes. These conclusions are supported in the present study, which proves in a quantitative way that storage method is the most significant predictor of loss risk.

This argument is also supported by the significance of storage period. According to the results, the longer the storage, the higher the risk of losses, which is also consistent with the work of [Ukwuaba et al., 2024], who show that prolonged storage is one of the main factors leading to losses in cucumber production. This implies that the quality of storage and the time goods are kept should both be managed to reduce spoilage.

Type of crop was also revealed to be a major aspect, with highly perishable crops like cabbage showing greater levels of losses. This observation aligns with [Oluoch et al., 2024], who state that leafy vegetables are especially susceptible to post-harvest degradation because of their high moisture level and fragility during handling. The difference in loss levels across different crops underscores the necessity to focus on crop-specific post-harvest methods.

Infrastructure differences are depicted in geographic variations in losses. The findings reveal

a stronger loss rate in the north and rural regions, consistent with [Pluato et al., 2024], who report post-harvest losses as being more severe in locations with low infrastructure, poor storage facilities, and inadequate transportation networks. These results highlight the significance of socio-economic and environmental circumstances in determining post-harvest outcomes.

The findings substantiate the assumption that post-harvest losses are the result of intertwined operational, environmental, and infrastructural factors. This reinforces the conceptual framework adopted in the literature review and emphasises the need to use integrated methods to reduce losses.

6.2.2 Predictive Accuracy of Machine Learning Models (RQ2)

The paper shows that structured agricultural data using simple machine learning models can successfully predict the risk of post-harvest losses. Random Forest was the top model among those tested, with Decision Tree coming in closely after, while Logistic Regression showed lower recall.

These results align with current studies in the field of machine learning in agriculture. [Jabed and Murad, 2024] note that ensemble techniques, including Random Forest, are especially useful in adopting difficult, non-linear associations in farm-based data. This claim is supported by the better outcome of the Random Forest in the present study, suggesting that post-harvest loss is determined by the combination of many interacting factors, which cannot be best explained by linear techniques.

In the same way, [Meghraoui et al., 2024] focus on the fact that tree-based models are very appropriate for agricultural prediction tasks as they can process heterogeneous data and capture non-linear interactions. Their high results for the Decision Tree and Random Forest models in this research affirm their appropriateness in predicting post-harvest loss risk.

The lower performance of Logistic Regression indicates the weakness of the linear methodology in complicated agricultural systems. Although logistic regression is interpretable, it can simplify variable relationships and hence lower predictive accuracy. This observation supports the essence of using models appropriate to the type of data.

The high accuracy achieved in this research can indicate that there exist high-quality predictor features, represented as storage method and loss-related variables. Although this improves model performance, concerns about data quality and overfitting are also raised. However, the validity of the results is enhanced by the use of several evaluation metrics and model comparison. In general, the results show that simple and understandable machine learning models can be highly predictive, even on small datasets.

6.2.3 Application of Predictions in Decision-Making (RQ3)

The important contribution that this study makes is to show that machine learning predictions can be useful in helping to make practical decisions in agribusiness. The findings indicate that predictive models have the ability to determine high-risk situations, which are dependent on storage approach and time, and therefore allow farmers to make advance actions.

This is in line with the concept of decision support systems (DSS) in the agricultural sector. [Duan et al., 2021] believe that DSS has the potential to increase agro productivity because it can help convert information into action. The given research builds upon that idea by demonstrating how machine learning models can serve as the core of such systems, giving risk forecasts that are used in decision-making.

The interpretability of Decision Trees adds further practical value. The model's rule-based format enables farmers and stakeholders to understand the circumstances under which a loss is probable. For example, storage duration thresholds revealed in the model can be applied in deciding optimum storage times. This proves how predictive models can be converted into rules that can be easily implemented in practice.

In addition, storage practices are identified as key factors for intervention. The risks of losses can be significantly minimized by improving storage methods or decreasing storage time. These lessons can be especially useful to smallholder farmers who require economical and feasible solutions. The output of such decision support systems is however determined by their access and use. According to [Duan et al., 2021], most existing systems are excessively complicated in the context of the smallholder. This research addresses this gap by concentrating on simple and interpretable models which can be applied using easily accessible data.

6.3 Theoretical Implications

This research contributes to the growing literature on machine learning in agriculture by demonstrating its applicability to post-harvest loss prediction. The study shows that machine learning can be used to address the problem of post-harvest management [Jabed and Murad, 2024, Meghraoui et al., 2024]. The results also support the criticality of incorporating data-driven strategies with conventional agricultural wisdom. The study offers a more systematic analysis of what drives loss after harvest by determining the strength of key drivers of losses. This helps in the creation of better theoretical models of agricultural systems.

6.4 Practical Implications

The applications of this study are highly practical. For smallholder farmers, the findings provide clear guidelines on how to reduce post-harvest losses. Better storage, shorter storage durations, and more efficient handling practices can play a great role in minimizing losses.

For policymakers, the findings demonstrate the importance of investment in storage facilities and training programmes. The geographic variations in loss rates can be reduced by addressing infrastructural differences between areas.

For agribusiness stakeholders, the research proves the importance of data-driven decision-making. Simple machine learning models can be implemented to improve supply chain efficiency and minimize losses along the way.

6.5 Limitations of the Study

Although it contributed within its confines, there are a few limitations of this study. Since it uses a dataset covering one country, the findings might be of limited generalisability to other settings. Also, the data is cross-sectional, which makes it impossible to analyze temporal changes in post-harvest losses. The high performance of the models can indicate the existence of strong predictive features, which increases the chances of overfitting. Moreover, the research centers on few variables, and additional factors including market forces and climate change may further enhance predictive ability.

6.6 Future Research Directions

Future study may build on this work by including bigger and more varied datasets, such as multi-country data. There would be an opportunity to include environmental variables (e.g. temperature, precipitation) that would benefit model performance. Besides, further examination of more sophisticated machine learning algorithms such as deep learning might yield more information, but should be weighed against the requirement of interpretability. The development and testing of real-time decision support systems would also help to close the gap between research and practice.

Chapter 7

Conclusion

7.1 Conclusion

The aim of this study was to design and test a machine learning-based model to predict the post-harvest loss risk among smallholder agribusinesses in developing nations, specifically Nigeria, based on LSMS-ISA data [World Bank, 2024]. This study focused on answering three major research questions: understanding the principal determinants of post-harvest losses, determining the predictive performance of simple machine learning models, and determining how those predictions can be used to inform viable agribusiness management decisions.

The results show that post-harvest losses are still a major menace, with a large percentage of smallholder farmers being affected. The analysis showed that storage-related variables, especially storage method and length of storage, are the most crucial factors of loss risk. Type of crop and geographic location are also important factors which introduce differences attributable to perishability and infrastructure facilities. These findings affirm the hypothesis that post-harvest losses are determined by a set of operational, environmental, and socio-economic factors.

Regarding predictive modelling, the researchers discovered that simple machine learning models are capable of attaining high levels of accuracy with structured tabular data. The best of the models analysed was the Random Forest, which showed a greater capacity to compute non-linear, complex relationships between variables. The Decision Tree also demonstrated high performance and presented useful interpretability, while Logistic Regression offered a strong baseline. These lessons show that resource-consuming and complex tools are not always necessary for effective predictive performance.

Significantly, the research revealed that machine learning predictions may be converted into agribusiness action insights. The high-risk situations can be determined, helping farmers to make the right choices about whether to store, handle, and sell products at the right time. This supports the development of low-cost decision support systems in smallholder scenarios, helping to develop efficiency and minimize losses. In general, the study fulfills its purpose, as it offers a feasible data-based model to forecast the risk of post-harvest losses. It adds to the academic and practical knowledge base, pointing to the possibilities of accessible machine learning solutions in solving critical agricultural problems.

7.2 Recommendations

This research presents a number of recommendations to farmers, policymakers, and future research bases, based on the findings of this study.

7.2.1 Recommendations for Smallholder Farmers

Smallholder farmers should first improve storage practices, since the method of storage and storage time were recognized as the most effective predictors of post-harvest losses. Even at the simplest level, adoption of better storage technologies can make a great difference in the reduction of spoilage. The farmers can also reduce storage time where possible, and should handle produce properly on harvesting and transportation as a way of minimising mechanical damage. Additionally, crop-specific measures must be embraced, especially for highly perishable crops that require careful handling.

7.2.2 Recommendations for Policymakers and Institutions

To manage structural aspects of post-harvest losses, policymakers ought to invest in rural infrastructure, especially storage and transport facilities. They should use training programmes to inform the farmers on best practices in post-harvest management. Additionally, it is necessary to facilitate the design and implementation of straightforward digital decision support systems that can offer real-time information to farmers using data. Agencies and other organizations in the agricultural sector should also enable access to agricultural data and encourage data-driven solutions. Enhancement of partnerships between the research and agriculture communities can bolster the incorporation of new agricultural solutions.

7.2.3 Recommendations for Future Research

Further investigation is recommended to enhance the scope of this study. Diversification of data is necessary since this study focuses on one country; future work should incorporate more extensive and intersecting databases covering multiple regions. Predictive accuracy could be further improved through the incorporation of environmental variables like temperature and rainfall. Moreover, the application of more sophisticated machine learning methods, including deep learning, could be explored while keeping interpretability and computational cost in mind. It is also necessary to build and test real-time decision support systems using the models presented in this research paper. This would help close the gap between theoretical and practical research, ensuring that predictive knowledge is converted into actionable outcomes.

7.3 Final Reflection

This paper has explained the mechanism by which agricultural knowledge and machine learning could help to solve one of the most pressing problems in the developing countries. The research demonstrates that much can be achieved with a strategic focus on simplicity, accessibility, and usability of solutions, and that effective technological tools do not necessarily require complex infrastructure. To sum up, the minimization of post-harvest losses is a challenge of both technical and social character, and it is a pathway toward better food security, the improved livelihoods of farmers, and the sustainable development of agriculture.

Appendix A

Supplementary Figures

This appendix contains supplementary output figures from the machine learning analysis, including additional model diagnostics, exploratory data analysis charts, and visualizations generated during the computational experiments.

```
# — 4.4 Causes of Loss ————————————————————————

cause_labels = {
    'phl_q4__1': 'Pest/Rodent',
    'phl_q4__2': 'Disease/Mould',
    'phl_q4__3': 'Moisture/Rain',
    'phl_q4__4': 'Poor Handling',
    'phl_q4__5': 'Transport',
    'phl_q4__6': 'Other'
}
existing_causes = {k: v for k, v in cause_labels.items() if k in df.columns}
if existing_causes:
    cause_df = df[list(existing_causes.keys())].sum().reset_index()
    cause_df.columns = ['code', 'count']
    cause_df['label'] = cause_df['code'].map(existing_causes)
```

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```
    cause_df = cause_df.sort_values('count', ascending=True)

    fig, ax = plt.subplots(figsize=(10, 6))
    bars = ax.barh(cause_df['label'], cause_df['count'],
                   color=PALETTE[:len(cause_df)], edgecolor='white', linewidth=1)
    ax.set_title('Reported Causes of Post-Harvest Loss', fontsize=14, fontweight='bold')
    ax.set_xlabel('Number of Occurrences')
    for bar, val in zip(bars, cause_df['count']):
        ax.text(bar.get_width() + 5, bar.get_y() + bar.get_height()/2,
                str(int(val)), va='center', fontsize=10)
    plt.tight_layout()
    plt.savefig('fig4_causes_of_loss.png', dpi=150, bbox_inches='tight')
    plt.show()
```

Figure A.1: Supplementary Figure 1

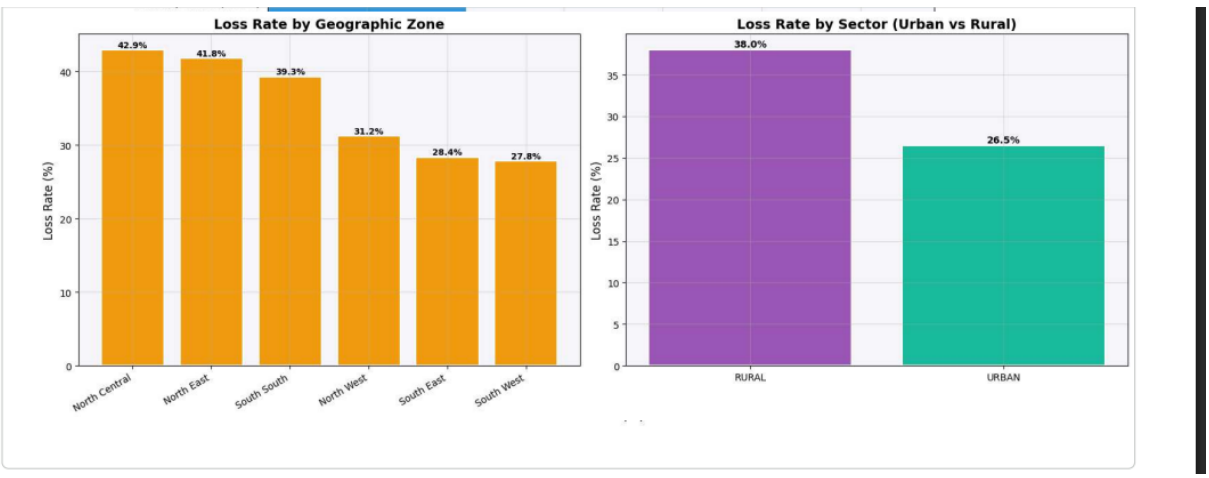

Figure A.2: Supplementary Figure 2

```
# ── 4.3 Loss by Zone and Sector ──────────────────────────

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# By zone
zone_loss = df.groupby('zone_clean')['loss_occurred'].mean().sort_values(ascending=False) * 100
axes[0].bar(zone_loss.index, zone_loss.values, color=PALETTE[3], edgecolor='white', linewidth=1.5)
axes[0].set_title('Loss Rate by Geographic Zone', fontweight='bold')
axes[0].set_ylabel('Loss Rate (%)')
axes[0].set_xticklabels(zone_loss.index, rotation=30, ha='right')
for i, (idx, val) in enumerate(zone_loss.items()):
    axes[0].text(i, val + 0.3, f'{val:.1f}%', ha='center', fontsize=9, fontweight='bold')

# By sector
sector_loss = df.groupby('sector_clean')['loss_occurred'].mean().sort_values(ascending=False) * 100
axes[1].bar(sector_loss.index, sector_loss.values,
            color=[PALETTE[4], PALETTE[5]][:len(sector_loss)], edgecolor='white', linewidth=1.5)
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 9/18

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```
axes[1].set_title('Loss Rate by Sector (Urban vs Rural)', fontweight='bold')
axes[1].set_ylabel('Loss Rate (%)')
for i, (idx, val) in enumerate(sector_loss.items()):
    axes[1].text(i, val + 0.3, f'{val:.1f}%', ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig('fig3_loss_by_zone_sector.png', dpi=150, bbox_inches='tight')
plt.show()
```

Figure A.3: Supplementary Figure 3

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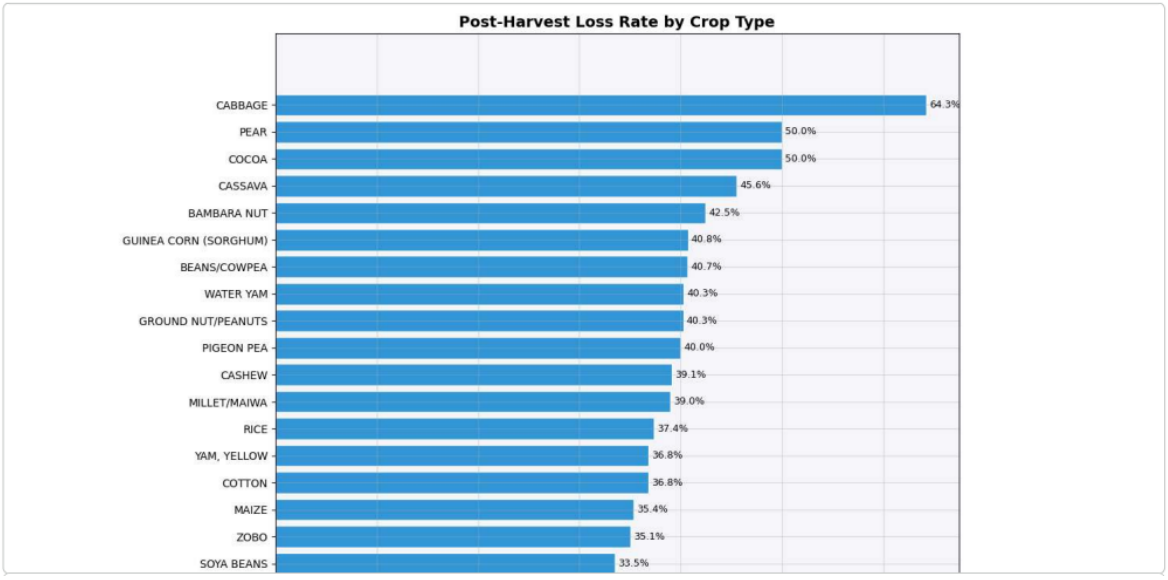

Figure A.4: Supplementary Figure 4

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```
# ── 4.2 Loss by Crop Type ──────────────────────────

if 'crop' in df.columns:
    crop_loss = df.groupby('crop')['loss_occurred'].agg(['mean', 'count']).reset_index()
    crop_loss.columns = ['crop', 'loss_rate', 'count']
    crop_loss = crop_loss[crop_loss['count'] >= 10].sort_values('loss_rate', ascending=True)
    crop_loss['loss_pct'] = crop_loss['loss_rate'] * 100

    fig, ax = plt.subplots(figsize=(12, max(6, len(crop_loss) * 0.4)))
    bars = ax.barh(crop_loss['crop'], crop_loss['loss_pct'],
                   color=PALETTE[2], edgecolor='white', linewidth=1)
    ax.set_xlabel('Loss Rate (%)')
    ax.set_title('Post-Harvest Loss Rate by Crop Type', fontsize=14, fontweight='bold')
    for bar, val in zip(bars, crop_loss['loss_pct']):
        ax.text(bar.get_width() + 0.3, bar.get_y() + bar.get_height()/2,
                f'{val:.1f}%', va='center', fontsize=9)
    plt.tight_layout()
    plt.savefig('fig2_loss_by_crop.png', dpi=150, bbox_inches='tight')
    plt.show()
```

Figure A.5: Supplementary Figure 5

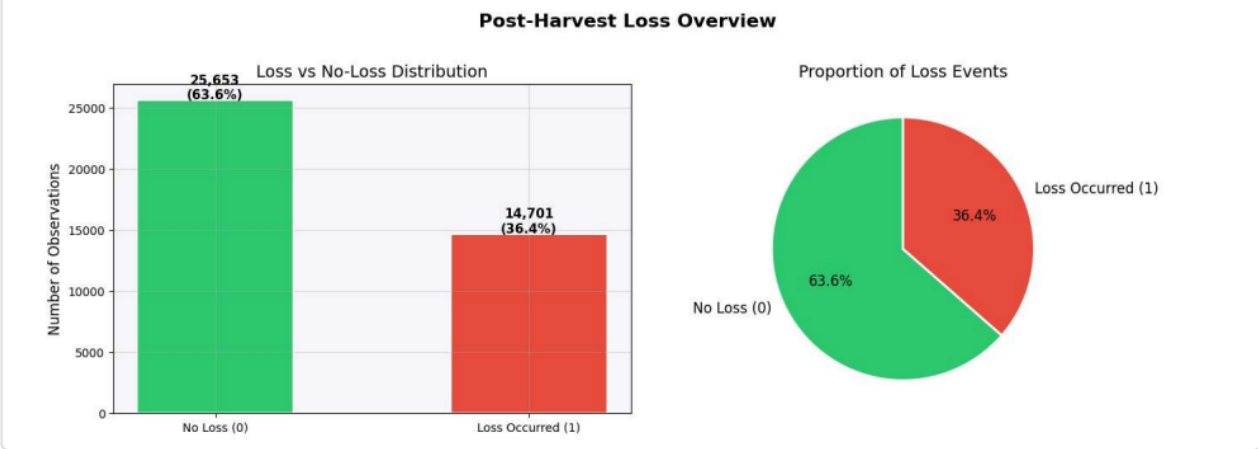

Figure A.6: Supplementary Figure 6

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```
✅ Libraries loaded successfully.
```

```
# — 4.1 Target Variable Distribution ————————————————

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('Post-Harvest Loss Overview', fontsize=16, fontweight='bold', y=1.02)

counts = df['loss_occurred'].value_counts()
labels = ['No Loss (0)', 'Loss Occurred (1)']
colors = [PALETTE[0], PALETTE[1]]
bars = axes[0].bar(labels, counts.values, color=colors, edgecolor='white', linewidth=1.5, width=0.5)
axes[0].set_title('Loss vs No-Loss Distribution')
axes[0].set_ylabel('Number of Observations')
for bar, val in zip(bars, counts.values):
    axes[0].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 50,
                 f'{val:,}\n({val/len(df)*100:.1f}%)', ha='center', fontsize=11, fontweight='bold')

axes[1].pie(counts.values, labels=labels, colors=colors, autopct='%1.1f%%',
            startangle=90, textprops={'fontsize': 12},
            wedgeprops={'edgecolor': 'white', 'linewidth': 2})
axes[1].set_title('Proportion of Loss Events')

plt.tight_layout()
plt.savefig('fig1_target_distribution.png', dpi=150, bbox_inches='tight')
plt.show()
```

Figure A.7: Supplementary Figure 7

```
# Set global plot style
plt.rcParams.update({
    'figure.facecolor': 'white',
    'axes.facecolor': '#f8f9fa',
    'axes.grid': True,
    'grid.alpha': 0.4,
    'font.family': 'DejaVu Sans',
    'axes.titlesize': 14,
    'axes.labelsize': 12,
})

PALETTE = ['#2ecc71', '#e74c3c', '#3498db', '#f39c12', '#9b59b6', '#1abc9c']
print("✅ Libraries loaded successfully.")
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 5/18

Figure A.8: Supplementary Figure 8

```
storage_duration     3.19    1.78    1.0     3.0     48.0
quantity_lost        3.98    1.07    0.5     4.0     16.0
qty_lost_1           3.02    1.23    1.0     3.0     30.0
qty_lost_2           2.11    0.65    1.0     2.0     14.0
value_lost_1      2106.10  930.94  200.0  2000.0  40000.0
loss_occurred        0.36    0.48    0.0     0.0      1.0

🌍 Loss Occurrence Rate by Geographic Zone (%):
loss_occurred      0      1
zone_clean
North Central  57.08  42.92
North East     58.20  41.80
North West     68.75  31.25
South East     71.65  28.35
South South    60.71  39.29
South West     72.17  27.83
```

```
# Categorical Frequency
print("\n🌍 Distribution by Geographic Zone:")
print(df['zone_clean'].value_counts())

print("\n🌾 Top 5 Crops in Dataset:")
print(df['crop'].value_counts().head(5))
```

```
🌍 Distribution by Geographic Zone:
zone_clean
North East       11554
North West       11131
North Central     7108
South East        4814
South South       3296
South West        2451
Name: count, dtype: int64

🌾 Top 5 Crops in Dataset:
crop
MAIZE                    8152
GUINEA CORN (SORGHUM)    5395
MILLET/MAIWA             3430
RICE                     3368
BEANS/COWPEA             3307
Name: count, dtype: int64
```

Figure A.9: Supplementary Figure 9

```
print("\n" + "="*60)
print("SECTION 4: EXPLORATORY DATA ANALYSIS")
print("="*60)
```

```
============================================================
SECTION 4: EXPLORATORY DATA ANALYSIS
============================================================
```

```
# Summary of numeric variables
stats_summary = df[num_cols + ['loss_occurred']].describe().T
print("\n📊 Summary Statistics for Key Variables:")
print(stats_summary[['mean', 'std', 'min', '50%', 'max']].round(2))

# Loss occurrence by Zone
zone_stats = pd.crosstab(df['zone_clean'], df['loss_occurred'], normalize='index') * 100
print("\n🌍 Loss Occurrence Rate by Geographic Zone (%):")
print(zone_stats.round(2))
```

```
📊 Summary Statistics for Key Variables:
                    mean     std    min     50%      max
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 4/18

Figure A.10: Supplementary Figure 10

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```
#missing values
# Fill cause flags with 0 (no loss = no cause)
flag_cols = [c for c in df.columns if 'phl_q4__' in c or 'phl_q07__' in c]
df[flag_cols] = df[flag_cols].fillna(0)

# Fill numeric with median
num_cols = ['storage_duration', 'quantity_lost', 'qty_lost_1', 'qty_lost_2', 'value_lost_1']
for col in num_cols:
    if col in df.columns:
        df[col] = df[col].fillna(df[col].median())
```

```
# Fill categorical with 'Unknown'
cat_cols = ['sector_clean', 'zone_clean', 'storage_method', 'storage_type_1', 'crop']
for col in cat_cols:
    if col in df.columns:
        df[col] = df[col].fillna('Unknown')
```

```
print(f"\nMissing values after cleaning:")
print(df[['loss_occurred', 'storage_duration', 'quantity_lost', 'sector_clean', 'zone_clean']].isnull().sum())
print("\n✅ Data cleaning complete.")
```

```
Missing values after cleaning:
loss_occurred       0
storage_duration    0
quantity_lost       0
sector_clean        0
zone_clean          0
dtype: int64

✅ Data cleaning complete.
```

Figure A.11: Supplementary Figure 11

```
# Cause flags from episode 1
cause2_cols = [c for c in df2.columns if 'phl_q07__' in c and c.endswith('_1')]
for col in cause2_cols:
    df2[col] = pd.to_numeric(df2[col], errors='coerce').fillna(0).astype(int)

print("\n✅ Both files cleaned.")
```

```
✅ Both files cleaned.
```

```
#3.3 Merge the two files

merge_keys = ['hhid', 'cropcode']

# Check merge keys exist in both
for key in merge_keys:
    assert key in df1.columns, f"Missing {key} in df1"
    assert key in df2.columns, f"Missing {key} in df2"

df = df1.merge(df2, on=merge_keys, how='left', suffixes=('_p1', '_p2'))

print(f"\nMerged dataset shape: {df.shape[0]:,} rows × {df.shape[1]} columns")
print(f"Rows after merge: {len(df):,}")
```

```
Merged dataset shape: 40,354 rows × 102 columns
Rows after merge: 40,354
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 3/18

Figure A.12: Supplementary Figure 12

```
# df_q5 = storage method
df1['storage_method'] = df1['phl_q5'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()

print(f"Target variable distribution:")
print(df1['loss_occurred'].value_counts())
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 2/18

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```
print(f"\nLoss rate: {df1['loss_occurred'].mean()*100:.1f}%")
```

```
Target variable distribution:
loss_occurred
0    25653
1    14701
Name: count, dtype: int64

Loss rate: 36.4%
```

```
# Keep only rows where at least one loss episode occurred
# Episode 1 storage type
df2['storage_type_1'] = df2['phl_q08__1'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()
df2['storage_type_2'] = df2['phl_q08__2'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()

# Loss stage (where in supply chain did loss happen)
df2['loss_stage_1'] = pd.to_numeric(df2['phl_q03_1'], errors='coerce')
df2['loss_stage_2'] = pd.to_numeric(df2['phl_q03_2'], errors='coerce')

# Quantity lost in each episode
df2['qty_lost_1'] = pd.to_numeric(df2['phl_q04_1'], errors='coerce')
df2['qty_lost_2'] = pd.to_numeric(df2['phl_q04_2'], errors='coerce')

# Value lost
df2['value_lost_1'] = pd.to_numeric(df2['phl_q06_1'], errors='coerce')
```

Figure A.13: Supplementary Figure 13

```
# Clean sector (urban/rural)
df1['sector_clean'] = df1['sector'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()

# Clean zone
df1['zone_clean'] = df1['zone'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()

# Clean state
df1['state_clean'] = df1['state'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()
```

```
# df_q2 = storage duration (weeks), df_q3 = quantity lost
df1['storage_duration'] = pd.to_numeric(df1['phl_q2'], errors='coerce')
df1['quantity_lost'] = pd.to_numeric(df1['phl_q3'], errors='coerce')

# df_q4__1 to df_q4__6 = cause flags (pest, disease, moisture, handling, transport, other)
cause_cols = [c for c in df1.columns if c.startswith('phl_q4__')]
for col in cause_cols:
    df1[col] = pd.to_numeric(df1[col], errors='coerce').fillna(0).astype(int)
```

```
# df_q5 = storage method
df1['storage_method'] = df1['phl_q5'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()

print(f"Target variable distribution:")
print(df1['loss_occurred'].value_counts())
```

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Figure A.14: Supplementary Figure 14

```
✅ sectaphl1 loaded: 40,354 rows × 19 columns
✅ sectaphl2 loaded: 7,002 rows × 70 columns

--- sectaphl1 columns ---
['zone', 'state', 'lga', 'sector', 'ea', 'hhid', 'indiv', 'cropcode', 'FILTER1', 'phl_q1', 'phl_q2', 'phl_q3', 'phl_q4__1', 'phl_q4__2', 'phl_q4__3', 'phl_q4__4', 'phl_q4__5', 'phl_q4__6',

--- sectaphl2 columns ---
['zone', 'state', 'lga', 'sector', 'ea', 'hhid', 'cropcode', 'phl_q02_1', 'phl_q03_1', 'phl_q04_1', 'phl_q05_1', 'phl_q06_1', 'phl_q07__1_1', 'phl_q07__2_1', 'phl_q07__3_1', 'phl_q07__4_1'
```

```
print("\n" + "="*60)
print("SECTION 3: DATA CLEANING")
print("="*60)
```

```
============================================================
SECTION 3: DATA CLEANING
============================================================
```

```
# Standardise column names to lowercase
df1.columns = df1.columns.str.lower().str.strip()
df2.columns = df2.columns.str.lower().str.strip()
```

```
# Extract clean crop name (remove numeric code prefix)
df1['crop'] = df1['cropcode'].astype(str).str.extract(r'\d+\.\s*(.+)')[0].str.strip()
```

```
# Create binary target variable: 1 = Loss occurred, 0 = No loss
df1['loss_occurred'] = df1['phl_q1'].astype(str).str.contains('1\. YES|1\.YES|YES', case=False, na=False).astype(int)
```

Figure A.15: Supplementary Figure 15

```
# Set global plot style
plt.rcParams.update({
    'figure.facecolor': 'white',
    'axes.facecolor': '#f8f9fa',
    'axes.grid': True,
    'grid.alpha': 0.4,
    'font.family': 'DejaVu Sans',
    'axes.titlesize': 14,
    'axes.labelsize': 12,
})
```

```
#load data
df1 = pd.read_csv('sectaphl1_harvestw5.csv')
df2 = pd.read_csv('sectaphl2_harvestw5.csv')
```

```
print(f"✅ sectaphl1 loaded: {df1.shape[0]:,} rows × {df1.shape[1]} columns")
print(f"✅ sectaphl2 loaded: {df2.shape[0]:,} rows × {df2.shape[1]} columns")
print("\n--- sectaphl1 columns ---")
print(list(df1.columns))
print("\n--- sectaphl2 columns ---")
print(list(df2.columns))
```


Figure A.16: Supplementary Figure 16


```
# !pip install imbalanced-learn -q
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKFold
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix, roc_curve, auc,
    ConfusionMatrixDisplay
)
from sklearn.inspection import permutation_importance

# Optional: for handling class imbalance
try:
    from imblearn.over_sampling import SMOTE
    SMOTE_AVAILABLE = True
except ImportError:
    SMOTE_AVAILABLE = False
    print("Note: imbalanced-learn not installed. SMOTE will be skipped.")
```

Figure A.17: Supplementary Figure 17

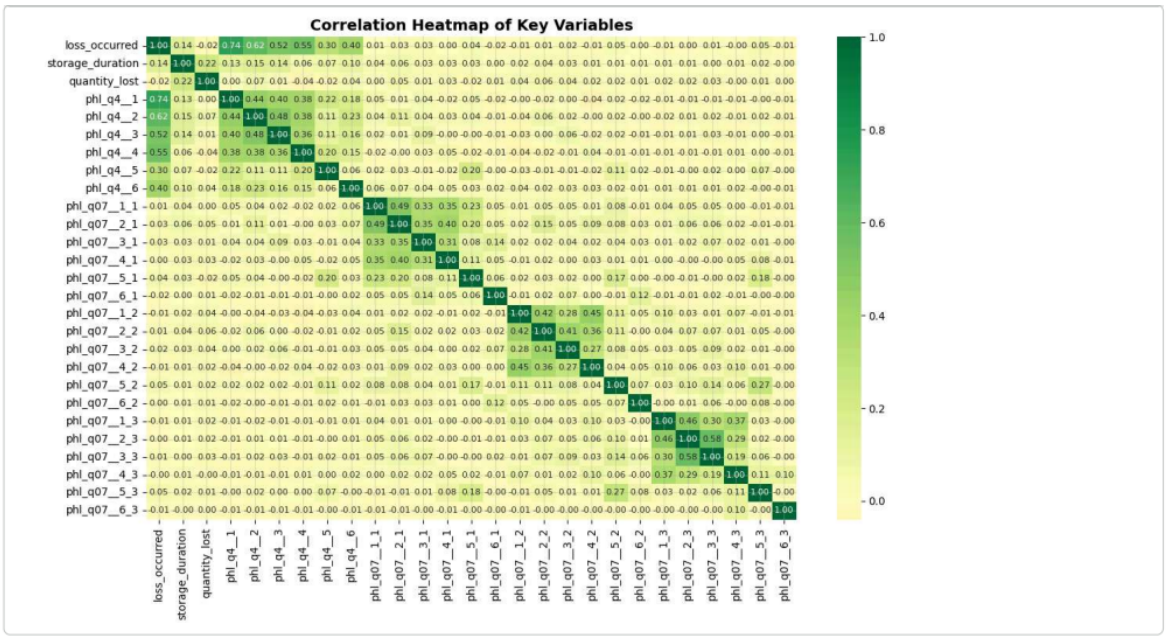

Figure A.18: Supplementary Figure 18

```
# ── 4.6 Correlation Heatmap ──────────────────────────────

numeric_df = df.select_dtypes(include=[np.number]).copy()
numeric_df = numeric_df.loc[:, numeric_df.std() > 0.01]
relevant_cols = ['loss_occurred', 'storage_duration', 'quantity_lost'] + \
                [c for c in numeric_df.columns if 'phl_q4' in c or 'phl_q07' in c]
relevant_cols = [c for c in relevant_cols if c in numeric_df.columns]

if len(relevant_cols) > 1:
    corr_matrix = numeric_df[relevant_cols].corr()
    fig, ax = plt.subplots(figsize=(12, 8))
    sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='RdYlGn', center=0, ax=ax, annot_kws={'size': 8})
    ax.set_title('Correlation Heatmap of Key Variables', fontsize=14, fontweight='bold')
    plt.tight_layout()
    plt.savefig('fig6_correlation_heatmap.png', dpi=150, bbox_inches='tight')
    plt.show()
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVEINJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 12/18

Figure A.19: Supplementary Figure 19

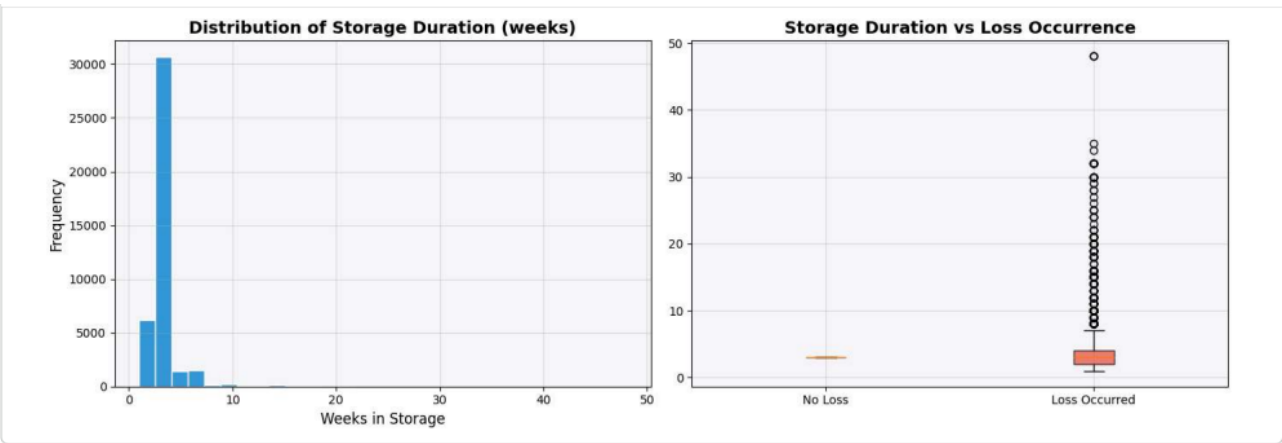

Figure A.20: Supplementary Figure 20

```
# ── 4.5 Storage Duration Distribution ──────────────────

if 'storage_duration' in df.columns:
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    df['storage_duration'].dropna().plot.hist(bins=30, ax=axes[0], color=PALETTE[2], edgecolor='white')
    axes[0].set_title('Distribution of Storage Duration (weeks)', fontweight='bold')
    axes[0].set_xlabel('Weeks in Storage')

    data_by_loss = [df[df['loss_occurred'] == 0]['storage_duration'].dropna(),
                    df[df['loss_occurred'] == 1]['storage_duration'].dropna()]
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVEINJfk6ZOqE4Rtqbxl#scrollTo=63RVTDvBqbjb&printMode=true 11/18

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```
    bp = axes[1].boxplot(data_by_loss, patch_artist=True, labels=['No Loss', 'Loss Occurred'])
    for patch, color in zip(bp['boxes'], [PALETTE[0], PALETTE[1]]):
        patch.set_facecolor(color)
        patch.set_alpha(0.7)
    axes[1].set_title('Storage Duration vs Loss Occurrence', fontweight='bold')
    plt.tight_layout()
    plt.savefig('fig5_storage_duration.png', dpi=150, bbox_inches='tight')
    plt.show()
```

Figure A.21: Supplementary Figure 21

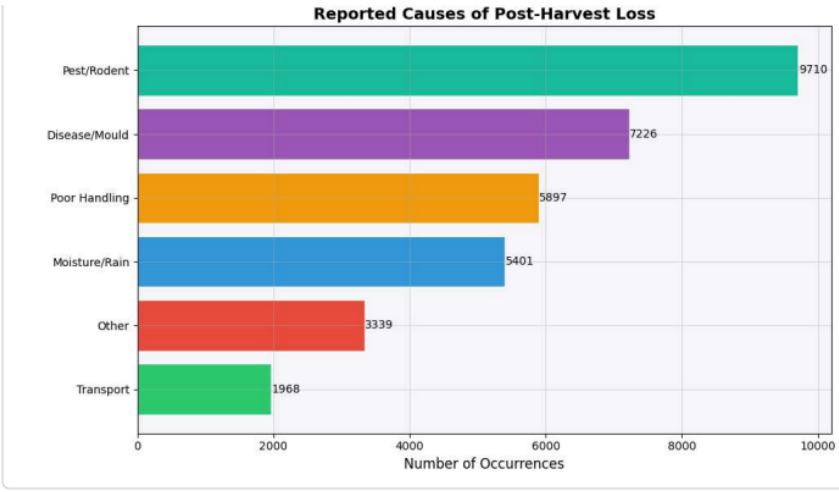

Figure A.22: Supplementary Figure 22

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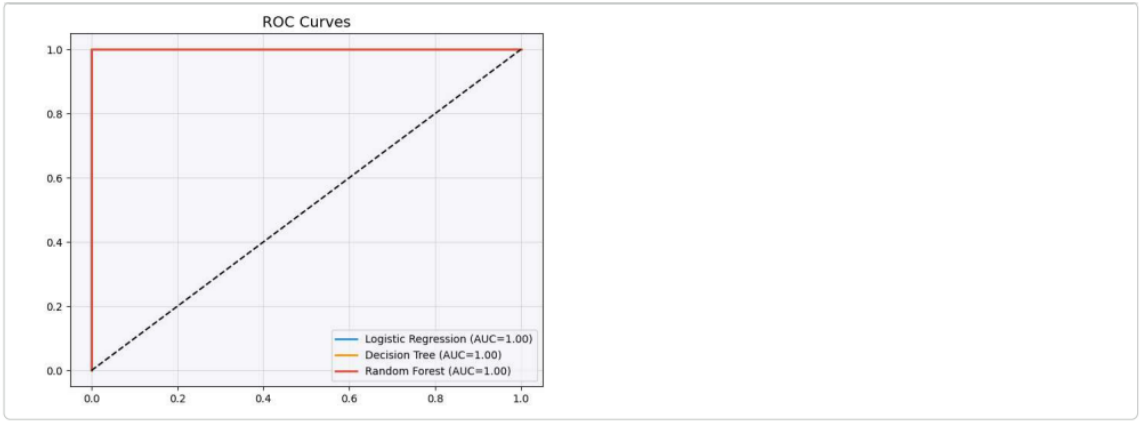

Figure A.23: Supplementary Figure 23

```
# ROC Curve
plt.figure(figsize=(8, 6))
for r, col in zip(results, [PALETTE[2], PALETTE[3], PALETTE[1]]):
    plt.plot(r['fpr'], r['tpr'], label=f"{r['Model']} (AUC={r['AUC']:.2f})", color=col, lw=2)
plt.plot([0, 1], [0, 1], 'k--')
plt.title('ROC Curves')
plt.legend()
plt.show()
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVEINJfk6ZOqE4Rtqbxl#scrollTo=63RVTDvBqbjb&printMode=true 16/18

Figure A.24: Supplementary Figure 24

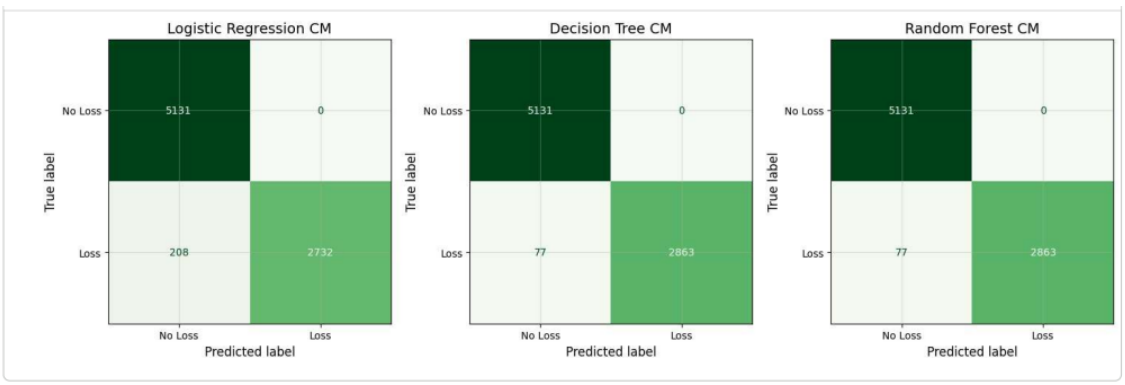

Figure A.25: Supplementary Figure 25

```
# Display Summary Table
metrics_df = pd.DataFrame([{k: v for k, v in r.items() if k not in ['fpr', 'tpr', 'pred']} for r in results])
print("\n📊 MODEL PERFORMANCE:")
print(metrics_df.round(4).to_string(index=False))
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 15/18

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```
📊 MODEL PERFORMANCE:
              Model  Accuracy  Precision  Recall  F1 Score  AUC-ROC
Logistic Regression    0.9742        1.0  0.9293    0.9633   0.9715
      Decision Tree    0.9905        1.0  0.9738    0.9867   0.9920
      Random Forest    0.9905        1.0  0.9738    0.9867   0.9938
```

```
# Confusion Matrices Visual
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
for i, r in enumerate(results):
    cm = confusion_matrix(y_test, r['pred'])
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['No Loss', 'Loss'])
    disp.plot(ax=axes[i], cmap='Greens', colorbar=False)
    axes[i].set_title(f"{r['Model']} CM")
plt.show()
```

Figure A.26: Supplementary Figure 26

```
print("✅ Models trained: Logistic Regression, Decision Tree, Random Forest.")
```

```
✅ Models trained: Logistic Regression, Decision Tree, Random Forest.
```

```
#model evaluation
def evaluate_student_model(name, model, xtest, ytest):
    y_pred = model.predict(xtest)
    y_prob = model.predict_proba(xtest)[:, 1]

    # Calculate Student Metrics (Accuracy, Precision, Recall, F1)
    acc = accuracy_score(ytest, y_pred)
    prec = precision_score(ytest, y_pred, zero_division=0)
    rec = recall_score(ytest, y_pred, zero_division=0)
    f1 = f1_score(ytest, y_pred, zero_division=0)

    fpr, tpr, _ = roc_curve(ytest, y_prob)
    roc_auc = auc(fpr, tpr)

    return {
        'Model': name, 'Accuracy': acc, 'Precision': prec,
        'Recall': rec, 'F1 Score': f1, 'AUC-ROC': roc_auc,
        'fpr': fpr, 'tpr': tpr, 'pred': y_pred
    }

results = [
    evaluate_student_model('Logistic Regression', lr, X_test_scaled, y_test),
    evaluate_student_model('Decision Tree', dt, X_test, y_test),
    evaluate_student_model('Random Forest', rf, X_test, y_test)
]
```

Figure A.27: Supplementary Figure 27

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```
# 2. Decision Tree (The Rule-Based Model)
dt = DecisionTreeClassifier(max_depth=5, min_samples_leaf=20,
                            random_state=42, class_weight='balanced')
dt.fit(X_train, y_train)
```

```
▾ DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=5,
                       min_samples_leaf=20, random_state=42)
```

```
# 3. Random Forest (The Ensemble Model)
rf = RandomForestClassifier(n_estimators=200, max_depth=8,
                            random_state=42, class_weight='balanced', n_jobs=-1)
rf.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier(class_weight='balanced', max_depth=8, n_estimators=200,
                       n_jobs=-1, random_state=42)
```

Figure A.28: Supplementary Figure 28

```
# Build final X and y
X = df[feature_cols].copy().fillna(0)
y = df['loss_occurred'].copy()

# Feature display mapping for the student's report
feature_display_names = {
    'zone_clean_enc': 'Geographic Zone',
    'sector_clean_enc': 'Sector (Urban/Rural)',
    'crop_enc': 'Crop Variety',
    'storage_method_enc': 'Storage Method Used',
    'storage_duration': 'Duration of Storage (Weeks)'
}
display_names = [feature_display_names.get(c, c) for c in feature_cols]

print(f"✅ Leakage fixed. Using {len(feature_cols)} logical predictors.")
print(f"Features: {display_names}")
```

```
✅ Leakage fixed. Using 5 logical predictors.
Features: ['Geographic Zone', 'Sector (Urban/Rural)', 'Crop Variety', 'Storage Method Used', 'Duration of Storage (Weeks)']
```

```
# SECTION 6: MODEL TRAINING (LR, DT, RF)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Standardize for Logistic Regression
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# 1. Logistic Regression (The Baseline)
lr = LogisticRegression(max_iter=1000, random_state=42, class_weight='balanced')
lr.fit(X_train_scaled, y_train)
```

```
▾ LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)
```

Figure A.29: Supplementary Figure 29

```
print("\n" + "="*60)
print("SECTION 5: FEATURE ENGINEERING")
print("="*60)
```

```
============================================================
SECTION 5: FEATURE ENGINEERING
============================================================
```

```
feature_cols = []

# 1. Geographic & Demographic Predictors
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 13/18

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```
cat_features = ['zone_clean', 'sector_clean', 'crop']
for col in cat_features:
    if col in df.columns:
        le = LabelEncoder()
        df[f'{col}_enc'] = le.fit_transform(df[col].astype(str))
        feature_cols.append(f'{col}_enc')

# 2. Storage Characteristics (A true predictor)
if 'storage_method' in df.columns:
    le_storage = LabelEncoder()
    df['storage_method_enc'] = le_storage.fit_transform(df['storage_method'].astype(str))
    feature_cols.append('storage_method_enc')

# 3. Temporal Predictors
if 'storage_duration' in df.columns:
    feature_cols.append('storage_duration')
```

Figure A.30: Supplementary Figure 30

```
#Feature Importance
rf_importance = pd.DataFrame({
    'Feature': display_names,
    'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=True)

plt.figure(figsize=(10, 6))
plt.barh(rf_importance['Feature'], rf_importance['Importance'], color=PALETTE[3])
plt.title('Feature Importance: Key Drivers of Post-Harvest Loss', fontsize=14, fontweight='bold')
plt.xlabel('Importance Score (Mean Decrease in Impurity)')
plt.tight_layout()
plt.savefig('fig11_true_importance.png')
plt.show()
```

https://colab.research.google.com/drive/1q6fnAhYiiXlbnVElNJfk6ZOqE4RtqbxI#scrollTo=63RVTDvBqbjb&printMode=true 17/18

Figure A.31: Supplementary Figure 31

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